Aviation incidents-project

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Project Overview

- The company is looking to expand and enter the aviation industry and they are interested in purchasing and operating both commercial and private aircraft.

  This project analyses the aviation accident data in order to uncover trends and insights that may aid in decision making.

- GOAL: To help the aviation division identify aircraft with the lowest risk using the historical accident data.

- Data Source: national Transportation Safety Board aviation accident records (1962-2023).Dataset from Kaggle.

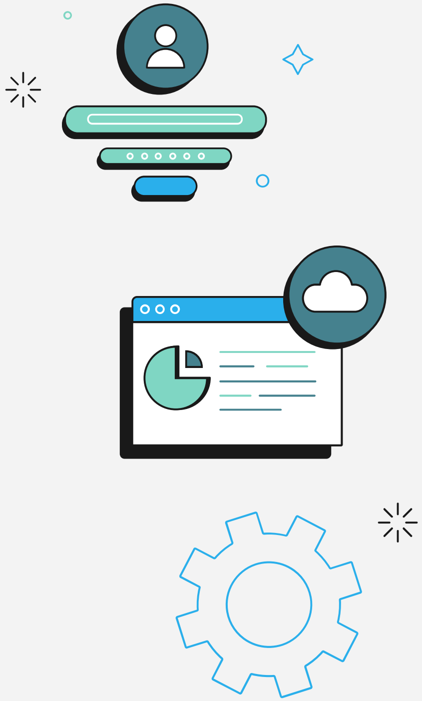
Business Understanding

- The company needs guidance on which aircraft pose the least risk
- Key question:

1. Which aircraft make and model present the lowest accidents and injury profiles?
2. What types of aircraft damages are most common and which models are most affected?
3. What are the safest aircraft makes and models based on the accident data?
4. Which aircraft should be prioritized for purchase to minimize operational risks and maximize passenger safety?

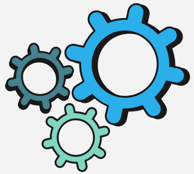
Data Understanding

Key variables used:



1. Aircraft type (Make and model)
2. Aircraft damage
3. Total injuries reports(serious, fatal, minor),
4. Event date
5. Injury severity.

Data preparation.



- What was done in data cleaning:

1. Cleaning and standardizing column names
2. handling missing values
3. Creating new features and columns (eg. total injury, aircraft type and injury ratio)

Data analysis

- A safety metrics table was created based on the data. It includes key measures like

1. total accidents : which was frequency of incidents. This was made from the Event ID
2. Fatalities: which was total fata injuries divided by the total accidents
3. injury severity : total injuries per incident
4. aircraft damage levels: percentage of aircraft destroyed or heavily damaged

- This helped standardize comparisons across aircraft types and narrow down safer options. Aircraft with strong safety records across these metrics were shortlisted as potentials.

Recommendation

- The riskiest plane to purchase from the metrics table is CESSNA 150M
- GRUMMAN-SCHWEIZER G-164A and CESSNA 150D stood out as top performers with
  - low fatal accident ratios.
  - low damage rates.
  - Minimal total fatalities.
- The number of injuries are reducing over time (years) this might be from the technological advancement or even improved safety standards, I would also advice to check on some current models.

Next Steps

- Explore additional data sources (costs, maintenance logs) to include financial and other operational feasibility
- Consult aviation experts for deeper insights
- Explore the impact of technological changes
- Add more recent data to update findings
- Evaluate cost and availability of recommended aircraft

Thank You

Question?

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